3. Z Transform

- *Reference*: Entire Chapter 3 of text.

- Recall that the Fourier transform (FT) of a DT signal $x[n]$ is

$$X\left(e^{jw}\right) = \sum_{k=-\infty}^{\infty} x[n]e^{-jwn}$$

In order for the FT to exist in the finite magnitude sense,

$$S = \sum_{k=-\infty}^{\infty} \left|x[n]\right| < \infty$$

As shown in Chapter 2, the FT does not always exist. If this occurs, it may be inconvenient to analyze the corresponding signal/system.

- In some cases, we can get around the problem by accepting the impulse function in the frequency domain as a valid function to represent the signal in the frequency domain.

- A more general, and very often more convenient, solution is to adopt the so-called *Z*-transform (ZT).

  As you will see in a moment, it is possible to obtain the ZT of signals whose FT do not exist.

The ZT can be viewed as a more general version of the FT. Its relationship with the FT is analogous to that between the Laplace transform and the Fourier transform of a continuous time signal.

3.1 Properties of Z-Transform

- The ZT of a discrete time signal $x[n]$ is

$$X(z) = \sum_{k=-\infty}^{\infty} x[k] z^{-k}$$

where z is a complex number. When we evaluate the ZT on the unit circle, i.e. we replace z in the above equation by

$$z = e^{jw} ,$$

then the FT of the signal is obtained (if it exists in the first place).

- **Example 1**: Determine the ZT of the right-sided exponential function

$$x[n] = a^n u[n] ,$$

where $u[n]$ is the unit-step function.

Solution:

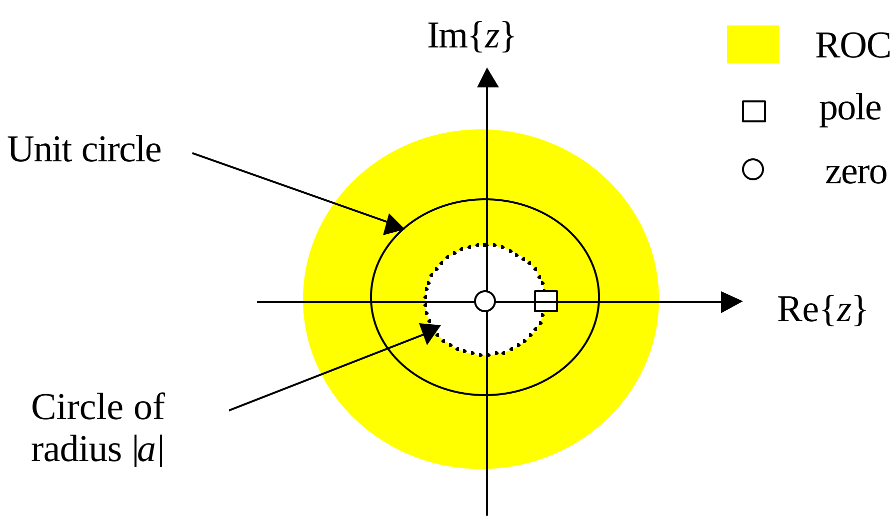
$$X(z) = \sum_{k=-\infty}^{\infty} x[k] z^{-k} = \sum_{k=0}^{\infty} a^k z^{-k} = \sum_{k=0}^{\infty} \left(az^{-1}\right)^k$$

If the above summation is to exist in the finite magnitude sense, then $\left|az^{-1}\right|<1$ or equivalently $|z|>|a|$. In this case

$$X(z)=\sum_{k=0}^{\infty}\left(az^{-1}\right)^{k}=\frac{1}{1-az^{-1}}=\frac{z}{z-a}; \qquad\qquad |z|>|a|$$

- In the above example, the region where the ZT exists, i.e. $|z|>|a|$, is called the *region of convergence* (ROC), the point $z=a$ where the ZT goes to infinity is called a *pole*, and the point $z=0$ where the ZT goes to zero is called a *zero*,

  The ROC and the pole-zero plot of Example 1 are shown below (assuming a is real and positive).



- **Example 2**: Determine the ZT of the left-sided signal

$$x[n]=-a^{n}u[-n-1]$$

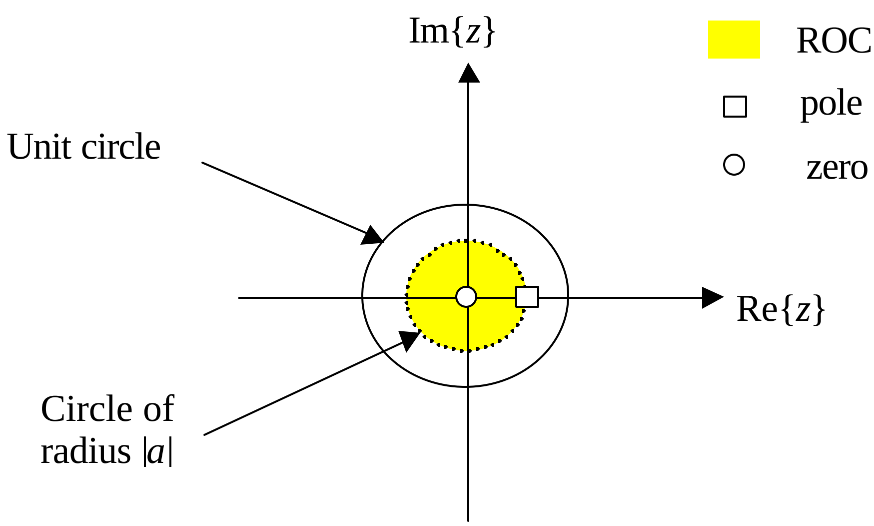
Solution:

$$X(z) = \sum_{k=-\infty}^{\infty} x[k] z^{-k} = -\sum_{k=-\infty}^{-1} a^k z^{-k} = -\sum_{m=1}^{\infty} \left(a^{-1}z\right)^m = 1 - \sum_{m=0}^{\infty} \left(a^{-1}z\right)^m$$

The sum exists only if $\left|a^{-1}z\right| < 1$ or $|z| < |a|$. In this case

$$X(z) = 1 - \sum_{m=0}^{\infty} \left(a^{-1}z\right)^m = 1 - \frac{1}{1 - a^{-1}z} = \frac{z}{z-a}; \qquad |z| < |a|$$

The answer is similar to that in Example 1, except for the ROC.



Through these two examples, it becomes evidient that it is very important to specify the ROC of the ZT.

- **Example 3**: Determine the ZT of

$$x[n] = a^n u[n] + b^n u[n]$$

Solution:

Using the result from Example 1 and the fact that the ZT is a linear operation, we have

$$X(z) = \frac{z}{z-a} + \frac{z}{z-b}$$

The first term exists if $|z| > |a|$ and the second term exists if $|z| > |b|$. So the ROC of $X(z)$ is the overlap of the individual ROCs, i.e. the exterior of a circle of radius

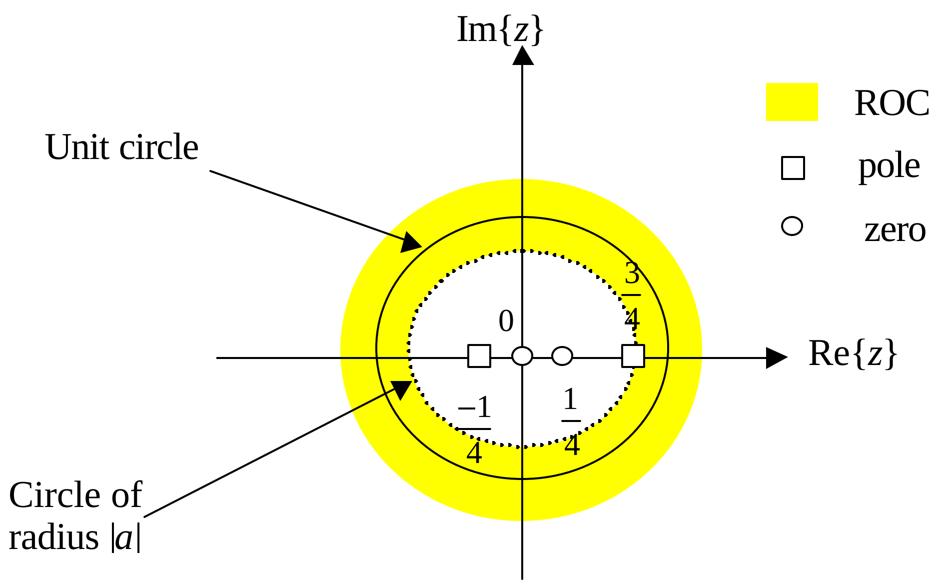
$$R = \max\{|a|, |b|\}$$

As for poles and zeros, we first rewrite $X(z)$ as

$$X(z) = \frac{z}{z-a} + \frac{z}{z-b} = \frac{2z\left(z - \frac{a+b}{2}\right)}{(z-a)(z-b)}; \qquad |z| > R$$

So the poles are located at $z = a$ and $z = b$; the zeros are located at $z = 0$ and $z = (a+b)/2$.

The ROC and the pole-zero plot for $a = 3/4,\ b = -1/4$ is shown below.



- **Example 4**: Determine the ZT of the 2-sided signal

$$x[n] = a^n u[n] - b^n u[-n-1]$$

Solution

Using results from both Examples 1 & 2, and the fact that ZT is a linear operation, we have

$$X(z) = \frac{z}{z-a} + \frac{z}{z-b}$$

The first term exists if $|z| > |a|$ while the second term exists if $|z| < |b|$. So the ROC of $X(z)$ is the overlap of the individual ROCs.

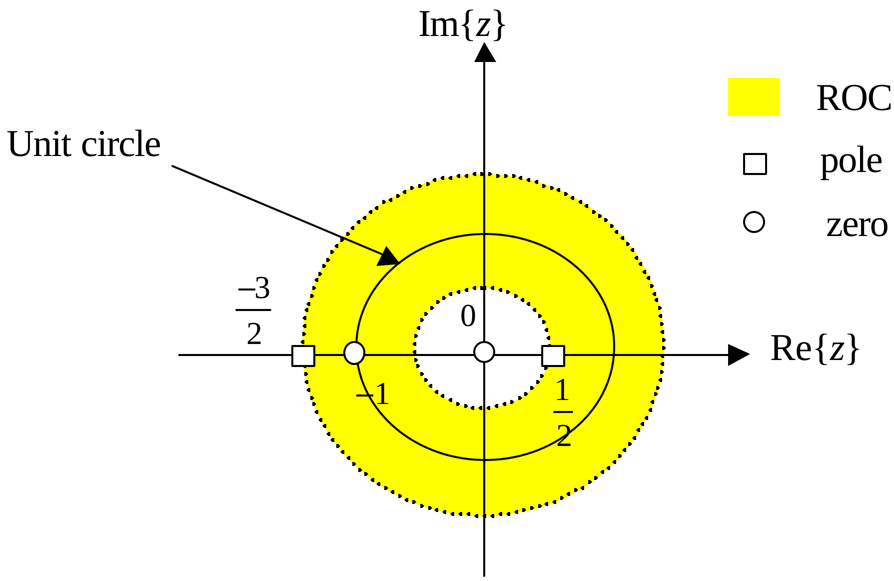
Case I: If $|a| < |b|$, then the ROC is the ring

$$|a| < |z| < |b|$$

In this case, the poles are located at $z = a$ and $z = b$, and the zeros are located at $z = 0$ and $z = (a+b)/2$; see Example 3.

Case II: If $|a| > |b|$, the ROC and hence $X(z)$ does not exist.

The ROC and pole-zero plot for $a = 1/2$, $b = -3/2$ is



- In summary, what the ZT does is that it describe a DT signal in terms of a set of poles $\{p_j\}$, a set of zeros $\{z_i\}$, and a ROC:

$$X(z) = \frac{\prod_{i=1}^{M}(z - z_i)^{a_i}}{\prod_{j=1}^{N}(z - p_j)^{b_j}}; \qquad z \in \text{ROC}$$

In the above equation, a_i and b_j represent respectively the multiplicity of the $i-$th zero and the $j-$th pole.

The FT, if exists, is obtained by evaluating the ZT on the unit circle. Specifically, the magnitude of the FT depends on the separations between the point e^{jw} and the poles and zeros according to

$$\left|X\left(e^{jw}\right)\right| = \frac{\prod_{i=1}^{M}\left|e^{jw} - z_i\right|^{a_i}}{\prod_{j=1}^{N}\left|e^{jw} - p_j\right|^{b_j}}; \qquad e^{jw} \in \text{ROC}$$

- **ROC of the ZT of a finite-duration signal:**

  - The signal $x[n]$ is of finite duration if $x[n]=0$ for $n<N_1$ and $n>N_2$, where both N_1 and N_2 are finite, with $N_1 \le N_2$.

  - In this case, the ZT is

$$X(z)=\sum_{n=N_1}^{N_2} x[n]z^{-n}$$

  - It can be easily shown that

$$\left|X(z)\right|=\left|\sum_{n=N_1}^{N_2} x[n]z^{-n}\right| \le \sum_{n=N_1}^{N_2} \left|x[n]\right|\left|z\right|^{-n} \le x_{\max}\sum_{n=N_1}^{N_2} \left|z\right|^{-n}$$

    where $x_{\max}=\max\left[\left|x[n]\right|\right]$. So the ZT is finite for any finite but non-zero value of z. If $N_2 \le 0$, then the ZT is also finite at $z=0$.

  - The ROC of a finite duration signal is the entire z-plane, excluding infinity and perhaps 0.

- **ROC of the ZT of a right-sided signal:**

  - By a right-sided signal, we mean $x[n]=0$ for $n<N_1$ for some finite N_1.

  - In this case, the ZT is

$$X(z)=\sum_{n=N_1}^{\infty} x[n]z^{-n}$$

- Let

$$q_0 = r_0 e^{jq}$$

represents a point on a circle of radius r_0 in the z-plane. If the ZT exists on this circle, then for any q

$$\left| X\left(q_o\right) \right| = \left| \sum_{n=N_1}^{\infty} x[n]\left(q_o\right)^{-n} \right| < \infty$$

- The ZT on the circle $q_1 = r_1 e^{jf}$, if exists, is

$$X\left(q_1\right) = \sum_{n=N_1}^{\infty} x[n]\left(q_1\right)^{-n} = \sum_{n=N_1}^{\infty} x[n]\left(q_0 \frac{q_1}{q_0}\right)^{-n} = \sum_{n=N_1}^{\infty} x[n]\left(q_0\right)^{-n} \, r^n$$

where

$$r = \frac{q_0}{q_1}$$

Define

$$g[n] = x[n]\left(q_0\right)^{-n} u[n - N_1],$$

$$f[n] = r^{-n} u[-n - N_1],$$

and

$$h[n] = g[n] \otimes f[n] = \sum_{k=-\infty}^{\infty} g[k] f[n-k].$$

Then it can be easily shown that

$$h[0] = X\left(q_1\right) = \frac{1}{2p}\int_{-p}^{p} G\left(e^{jw}\right) F\left(e^{jw}\right) dw$$

where $G\left(e^{jw}\right)$ and $F\left(e^{jw}\right)$ are the FTs of $g[n]$ and $f[n]$. It is interesting to point out that $G\left(e^{jw}\right)\big|_{w=0} = X\left(q_0\right)$, and in general

$$G\left(e^{jw}\right) = X\left(q_0 e^{jw}\right).$$

The important thing is that $G\left(e^{jw}\right)$ exists. The FT $F\left(e^{jw}\right)$ will also exist if $|r| < 1$, i.e. when $|r_1| > |r_0|$. If this condition holds, then the ZT on the circle of radius r_1 exists.

In summary, the ROC of a right-sided signal is to the exterior of a circle. The radius of this circle equals the largest pole (in magnitude sense).

- **ROC of the ZT of a left-sided signal:**

  - By a left-sided signal, we mean $x[n] = 0$ for $n > N_2$ for some finite N_2.

  - The ROC is to the interior of a circle. The radius of the circle equals the the smallest pole (in the magnitude sense).

- **ROC of the ZT of a two-sided signal:**

  - By a 2-sided signal, we mean a $x[n]$ that stretches from $n = -\infty$ to $n = \infty$.

- Since a 2-sided signal can be treated as the sum of a left sided signal with a ROC of $|z| < R_-$ and a right-sided signal with a ROC of $|z| > R_+$, its ROC is the overlap of the two.

- If $R_- > R_+$, the ROC is

$$R_+ < |z| < R_+ .$$

Otherwise, the ZT does not exist.

- **Exercise**: Show that the ROC must be a continuous region.

- **Exercise**: In this exercise, we use the notation

$$x[n] \xleftrightarrow{Z} X(z); \qquad \text{ROC: } R_x$$

to represent the signal $x[n]$ and its ZT $X(z)$ whose ROC is R_x. Show the following

(a) $ax_1[n] + bx_2[n] \xleftrightarrow{Z} aX_1(z) + bX_2(z);$ ROC: $R_{x_1} \cap R_{x_2}$

where $\cap$ represents intersection of two ROCs.

(b) $x[n - n_o] \xleftrightarrow{Z} z^{-n_0} X(z);$ ROC: R_x (except

for the possible additon or delection of $z = 0$, or $z = \infty$)

(c) $z_0^n x[n] \xleftrightarrow{Z} X(z / z_0);$ ROC: $|z_0| R_x$

(d) $nx[n] \xleftrightarrow{Z} -z\dfrac{dX(z)}{dz}$; ROC: R_x

(e) $x^*[n] \xleftrightarrow{Z} X^*(z^*)$; ROC: R_x

(f) $x[-n] \xleftrightarrow{Z} X^*(1/z^*)$; ROC: $1/R_x$

(g) $x_1[n] \otimes x_2[n] \xleftrightarrow{Z} X_1(z)X_2(z)$; ROC: $R_{x_1} \cap R_{x_2}$

3.2 Inverse Fourier Transform

- There are a number of approaches to determine a signal from its ZT.

3.2.1: By Inspection

- A ZT (or a component of a ZT) that you will encounter from time to time is

$$X(z) = \frac{1}{1-az^{-1}}; \qquad \text{ROC: } |z| > a$$

This is because many ZTs express the signal in terms of poles (and zeros).

- According to an earlier example, the time domain signal is $x[n] = a^n u[n]$.

- **Exercise**: Deteremine $x[n]$ if

$$X(z) = \frac{z^{-d}}{1-az^{-1}}; \qquad \text{ROC: } |z| > a$$

Answer: This is simply the signal $x_1[n] = a^n u[n]$ delayed by d samples (see Part (b) of the last exercise in Section 3.1). Consequently

$$x[n] = a^{(n-d)} u[n-d]$$

3.2.1 Power Series Expansion

- In its most primitive form, the ZT is a power series in z^{-1}, with the coefficient associated with $\left(z^{-1}\right)^n$ being $x[n]$.

- Consequently, the coefficients in the power series representation of the ZT give you the signal itself.

- **Example**: Determine $x[n]$ when

$$X(z) = e^z; \qquad\qquad |z| < \infty$$

Solution: The power series representation of $X(z)$ is

$$X(z) = e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = \sum_{m=-\infty}^{0} \frac{z^{-m}}{(-m)!}; \qquad\qquad |z| < \infty$$

where $n!$ denotes n factorial. From this expression, we can deduce that

$$x[n] = \frac{1}{(-n)!} u[-n]$$

3.3.3 Partial Fraction Expansion

- We assume that the ZT is given in the form

$$X(z) = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M} \left(1 - c_k z^{-1}\right)}{\prod_{i=1}^{N} \left(1 - d_i z^{-1}\right)}$$

where $c_k,\ k = 1,2,\ldots,M$ are the zeros, $d_i,\ i = 1,2,\ldots,N$ are the poles, and a_0, b_0 are constants.

- The idea behind the Partial Fraction Expansion technique is to express the above ZT as a weighted sum of terms of the form

$$\frac{1}{\left(1 - d_i z^{-1}\right)^m}$$

where m is an integer. The inverse ZT of the above can be determined analytically. Specifically, let

$$f_m[n] \xleftrightarrow{Z} F_m(z) = \frac{1}{\left(1 - d z^{-1}\right)^m}$$

where d represents in general a pole of $X(z)$. Since

$$F_m(z) = -\frac{1}{d(m-1)} z^2 \frac{d}{dz} F_{m-1}(z)$$
$$= \frac{1}{d(m-1)} z\left(-z\frac{d}{dz} F_{m-1}(z) \right),$$

this means

$$f_m[n] = \frac{1}{d(m-1)} (n+1)\ f_{m-1}[n+1];$$

refer to (b) and (d) in the last exercise in Section 3.1. Since

$$f_1[n] = d^n u[n],$$

so

$$f_2[n] = (n+1) d^n u[n+1],$$

$$f_3[n] = \frac{1}{2}(n+2) d^n u[n+2],$$

$$f_4[n] = \frac{1}{3\times 2}(n+3) d^n u[n+3],$$

or in general

$$f_m[n] = \frac{1}{(m-1)!}(n+m-1) d^n u[n+m-1]$$

- To describe the partial fraction expansion technique, we consider three cases,

  (a) the number of zeros is less than the number of poles and there are no multiple-order poles,
  (b) the number of zeros is greater than the number of poles but still there are no multiple-order poles, and
  (c) there are more zeros than poles and there is one multiple-order pole.

  Comments:

  - A pole is of multiple-order if it is repeated more than 1 time.

  - The scenario of more than one multiple-order poles can be considered as a straightforward extension of case (c) above.

<u>Case 1 – $M < N$ and no multiple-order pole</u>

- All the N poles are different.

- In this case, the original ZT can be expressed as

$$X(z) = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M}\left(1 - c_k z^{-1}\right)}{\prod_{i=1}^{N}\left(1 - d_i z^{-1}\right)} = \sum_{i=1}^{N} \frac{A_i}{1 - d_i z^{-1}}$$

$$= \frac{A_1}{1 - d_1 z^{-1}} + \frac{A_2}{1 - d_2 z^{-1}} + \ldots + \frac{A_N}{1 - d_N z^{-1}}$$

Consequently

$$x[n]=\sum_{i=1}^{N} A_i\left(d_i\right)^n u[n]$$

The question that remains to be answered is: what are the A_i s?

- If we multiply both sides of $X(z)$ by $1-d_1z^{-1}$, the result is

$$\left(1-d_1z^{-1}\right)X(z)=A_1+\frac{A_2\left(1-d_1z^{-1}\right)}{1-d_2z^{-1}}+\ldots+\frac{A_N\left(1-d_1z^{-1}\right)}{1-d_Nz^{-1}}$$

Now, if we evalue the above at $z=d_1$, we simply have

$$\left[\left(1-d_1z^{-1}\right)X(z)\right]_{z=d_1}=A_1$$

- In general,

$$\left[\left(1-d_iz^{-1}\right)X(z)\right]_{z=d_i}=A_i$$

- **Example:** A left-sided signal has a ZT of

$$X(z)=\frac{1}{\left(1-\frac{1}{4}z^{-1}\right)\left(1-\frac{1}{2}z^{-1}\right)}$$

Determine the ROC and the inverse ZT.

Solution:

The ZT can be expressed as

$$X(z) = \frac{1}{\left(1-\frac{1}{4}z^{-1}\right)\left(1-\frac{1}{2}z^{-1}\right)} = \frac{A_1}{\left(1-\frac{1}{4}z^{-1}\right)} + \frac{A_2}{\left(1-\frac{1}{2}z^{-1}\right)}$$

where

$$A_1 = \left[\left(1-\tfrac{1}{4}z^{-1}\right)X(z)\right]_{z=1/4} = -1$$

and

$$A_2 = \left[\left(1-\tfrac{1}{2}z^{-1}\right)X(z)\right]_{z=1/2} = 2$$

Consequently

$$X(z) = \frac{-1}{\left(1-\frac{1}{4}z^{-1}\right)} + \frac{2}{\left(1-\frac{1}{2}z^{-1}\right)}$$

Given that $x[n]$ is a left-sided signal, the first term must have a region of convergence of $|z|<1/4$ and the second term must have a region of convergence of $|z|<1/2$. The overall ROC is the overlap of the two, which is simply $|z|<1/4$.

The inverse ZTs of the first and second terms are $(1/4)^n u[-n-1]$ and $-2(1/2)^n u[-n-1]$ respectively. So

$$x[n] = (1/4)^n u[-n-1] - 2(1/2)^n u[-n-1]$$
$$= \left\{(1/4)^n - 2(1/2)^n\right\} u[-n-1] \quad .$$

- **Exercise**: Repeat the last example for a 2-sided signal.

<u>Case 2 – M greater than or equal to N and no multiple-order pole</u>

- As in Case 1, all the poles are different.

- In this case, the original ZT can be expressed as

$$X(z) = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M}\left(1 - c_k z^{-1}\right)}{\prod_{i=1}^{N}\left(1 - d_i z^{-1}\right)} = Q(z) + \frac{R(z)}{\prod_{i=1}^{N}\left(1 - d_i z^{-1}\right)}$$

where

$$Q(z) = \sum_{r=0}^{M-N} B_r z^{-r}$$

and $R(z)$ are respectively the quotient and remainder polynomials (in z^{-1}) when $b_0 / a_0 \prod_{k=1}^{M}\left(1 - c_k z^{-1}\right)$ is divided by $\prod_{i=1}^{N}\left(1 - d_i z^{-1}\right)$.

Since the degree of the remainder polynomial (in z^{-1}) is less than N, it means the second term in $X(z)$ can be expressed in the same form as that in Case 1. Consequently, we can rewrite $X(z)$ as

$$X(z) = Q(z) + \frac{R(z)}{\prod_{i=1}^{N}\left(1 - d_i z^{-1}\right)}$$

$$= \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{i=1}^{N} \frac{A_i}{\left(1 - d_i z^{-1}\right)},$$

where again

$$A_i = \left[\left(1 - d_i z^{-1}\right) X(z)\right]_{z=d_i}$$

- **Example:** Determine the IZT of

$$X(z) = \frac{-1 + \frac{1}{2} z^{-1} - \frac{1}{4} z^{-2} + \frac{1}{16} z^{-3}}{\left(1 - \frac{1}{4} z^{-1}\right)\left(1 - \frac{1}{2} z^{-1}\right)}; \qquad |z| < \frac{1}{4}$$

Solution:

- From the ROC, we know that this is a left-sided signal.
- The denominator can be written as $D(z) = 1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2}$.
- Dividing the numerator $C(z) = -1 + \frac{1}{2} z^{-1} - \frac{1}{4} z^{-2} + \frac{1}{16} z^{-3}$ by the denominator, using long division, yields

$$Q(z) = 1 + \frac{1}{2} z^{-1}$$

and

$$R(z) = -2 + \tfrac{3}{4} z^{-1}.$$

This means

$$X(z) = 1 + \tfrac{1}{2} z^{-1} + \frac{-2 + \tfrac{3}{4} z^{-1}}{(1 - \tfrac{1}{4} z^{-1})(1 - \tfrac{1}{2} z^{-1})}$$

$$= 1 + \tfrac{1}{2} z^{-1} + \frac{A_1}{(1 - \tfrac{1}{4} z^{-1})} + \frac{A_2}{(1 - \tfrac{1}{2} z^{-1})},$$

where

$$A_1 = \left[\frac{-1 + \tfrac{1}{2} z^{-1} - \tfrac{1}{4} z^{-2} + \tfrac{1}{16} z^{-3}}{(1 - \tfrac{1}{4} z^{-1})(1 - \tfrac{1}{2} z^{-1})} (1 - \tfrac{1}{4} z^{-1}) \right]_{z=1/4} = -1$$

$$A_2 = \left[\frac{-1 + \tfrac{1}{2} z^{-1} - \tfrac{1}{4} z^{-2} + \tfrac{1}{16} z^{-3}}{(1 - \tfrac{1}{4} z^{-1})(1 - \tfrac{1}{2} z^{-1})} (1 - \tfrac{1}{2} z^{-1}) \right]_{z=1/2} = -1$$

- The inverse ZT is thus

$$x[n] = d[n] + \tfrac{1}{2} d[n-1] + \left(\tfrac{1}{4}\right)^n u[-n-1] + \left(\tfrac{1}{2}\right)^n u[-n-1]$$

<u>Case 3 – M greater than or equal to N and 1 multiple-order pole</u>

- The pole d_1 is repeated s times, i.e.

$$d_1 = d_2 = \cdots = d_s \,.$$

Each of the remaining $N-s$ poles has an order of 1. This means the ZT transform, in product form, is

$$X(z) = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M}\left(1-c_k z^{-1}\right)}{\left(1-p_1\right)^s \prod_{i=s+1}^{N}\left(1-d_i z^{-1}\right)}$$

- The ZT, in summation form, is

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{i=1}^{N} \frac{A_i}{\left(1-d_i z^{-1}\right)} + \sum_{m=1}^{s} \frac{C_m}{\left(1-d_1 z^{-1}\right)^m}$$

The determination of the A_i s and the B_r s are the same as before, i.e

$$A_i = \left[\left(1-d_i z^{-1}\right) X(z)\right]_{z=d_i} \; ; \qquad\qquad i = s+1, s+2, \ldots, N \,,$$

and B_r is the r-th coefficients of the quotient polynomial when the numerator $\left(b_0/a_0\right)\prod_{k=1}^{M}\left(1-c_k z^{-1}\right)$ of $X(z)$ is divided by its denominator $\left(1-p_1\right)^s \prod_{i=s+1}^{N}\left(1-d_i z^{-1}\right)$.

- The C_m s can be obtained by first replacing the z^{-1} by w and then compute

$$C_m = \frac{1}{(s-m)!(-d_1)^{s-m}} \left[\frac{d^{s-m}}{dw^{s-m}} \left\{ (1-d_1 w)^s X\left(w^{-1}\right) \right\} \right]_{w=d_1^{-1}} .$$

Note that

$$\begin{aligned}(1-d_1w)^s X\left(w^{-1}\right) &= (1-d_1w)^s \left\{ \sum_{r=0}^{M-N} B_r w^r + \sum_{i=1}^{N} \frac{A_i}{(1-d_i w)} + \sum_{m=1}^{s} \frac{C_m}{(1-d_1 w)^m} \right\} \\ &= (1-d_1w)^s \sum_{r=0}^{M-N} B_r w^r + (1-d_1w)^s \sum_{i=1}^{N} \frac{A_i}{(1-d_i w)} \\ &\quad + \sum_{r=1}^{s} C_r (1-d_1 w)^{s-r}\end{aligned}$$

Comments:

- Evaluating the $(s-m)$-th order derivatives of the first two terms at $w = d_1^{-1}$ will always yield zero.

- Evaluating the $(s-m)$-th order derivative of $C_r(1-d_1w)^{s-r}$ at $w = d_1^{-1}$ will be zero except when $r = m$. Note that when $r > m$, we will at some point end up with differentiating a constant. On the other hand when $r < m$, the result after differentiation will include power of the term $(1-d_1w)$, which when evaluated at $w = d_1^{-1}$ equal to zero.